

YOURSTORY



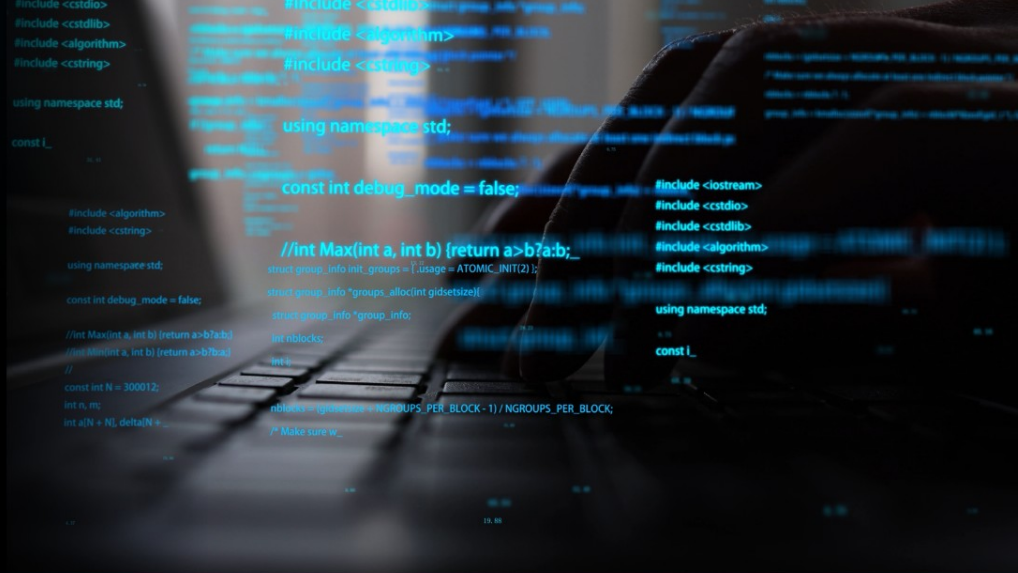
H2S



Coco CLI

HACKATHON

GCC Edition



Team Name : **Provenance**

Team Leader Name : **Manoj T**

Team Size : **2**

Problem Statement : **1. Risk, Fraud and Regulatory Intelligence Copilot**

The problem

Risk, Fraud and Regulatory Intelligence Copilot · banking and NBFC

Finding the risk signal is cheap. **Defending the number is expensive.**

TODAY

Ten minutes to find the signal. Two days to assemble an evidence pack an auditor will accept.

THE GAP

Ask two analysts the same question and they compute it differently. Booking or value date. Reversals in or out.

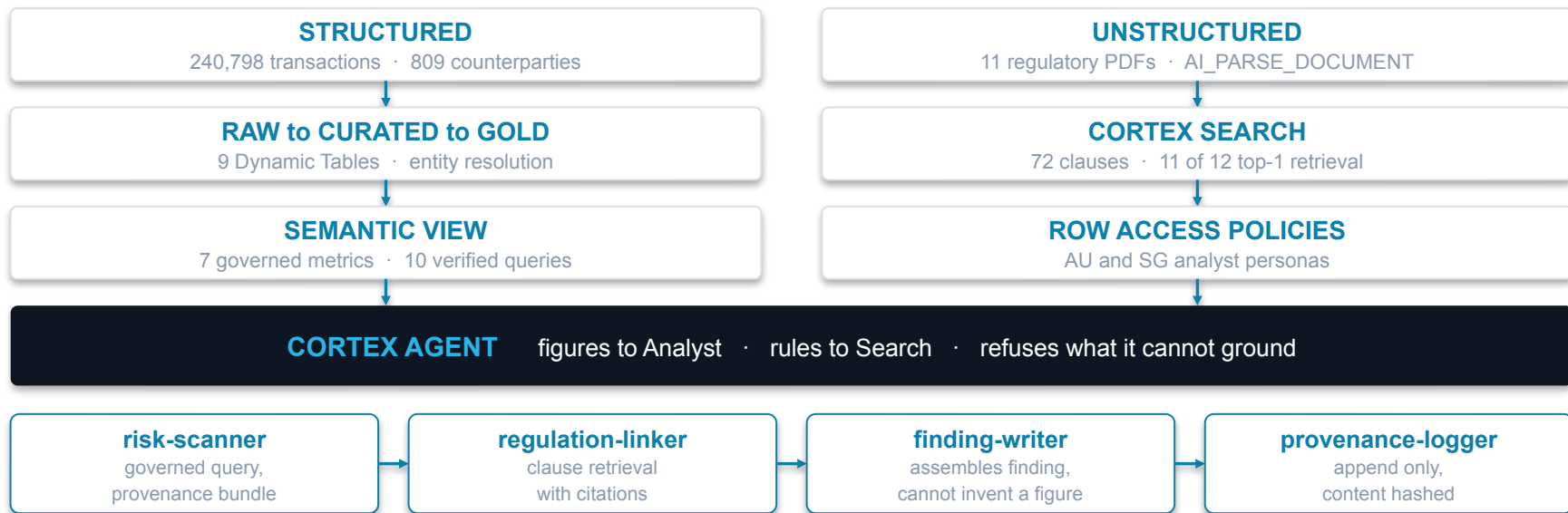
THE USER

A compliance analyst or MLRO at a mid-sized bank who has to put their name on the number.

PaperTrail does not answer with a number. It answers with a filing, in which every figure is a footnote that resolves to its governed definition, the exact SQL, the source rows, and the clause that makes it matter.

Architecture

Snowflake-native, end to end. No LangChain, no external vector store, no third-party LLM API.



Measured, not asserted

Governed against ungoverned. Same data, same model, full raw DDL, no hint of any governance rule.

Our hypothesis was wrong. An ungoverned model is not inconsistent. Across 25 runs the SQL was byte identical. It is perfectly confident, and wrong on 2 of 5.

QUESTION	UNGOVERNED	GOVERNED	
Suspicious volume, Q3 2026	8,797,923	4,295,229	+105%
Exposure to CP-STRUCT-01	87,185	87,185	agree
Alert closure rate, 2026	63.6%	63.6%	agree
SAR filing share	6.5%	6.5%	agree
High risk counterparties, Singapore	23	36	-36%

The \$4.5M overstatement comes from four missed decisions at once: booking date rather than value date, reversals counted, unsettled counted, and transactions on already closed alerts treated as suspicious. Exposure agrees only by coincidence, because that counterparty has no beneficial ownership links.

Measured outcomes

Scored against an answer key that was never loaded into Snowflake.

0.944

detection recall

5 of 6

typologies at F1 1.000

11 / 12

clause retrieval, top-1

5 / 5

agent tool routing

WHAT IT SAVES

The evidence pack is the artefact that takes an analyst two days by hand. PaperTrail emits it with the answer, and the figure re-derives from its own footnote in about 300 ms.

WHY NOT 100%

Velocity spike detection is weak. Removing it would take precision and F1 to 1.000. We kept it because the regulation requires it. A perfect score on self-authored data would be evidence of tuning.

Live app papertrail-provenance.vercel.app opens for anyone, no login

Repository github.com/tjmanoj/papertrail

Also deployed natively as **Streamlit in Snowflake**

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1125

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THANK YOU



```
#include <algorithm>
#include <string>

using namespace std;

const int debug_mode = false;

#include <algorithm>
#include <string>

using namespace std;

const int debug_mode = false;

//int Max(int a, int b) {return a>b?a:b;}
//int Min(int a, int b) {return a<b?a:b;}
//
const int N = 300012;
int n, m;
int a[N + N], delta[N + N];

// Make sure w
```